higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 2

(12041002)

18 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 8 pages and a formula sheet of 2 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only blue or black ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 Calculate the weight of an object having a mass of 100 kg. (2)

1.2 A vehicle travels at 60 km/h on a horizontal road.

Calculate the speed of the vehicle in m/s. (2)

1.3 Calculate the work done to push a car with a force of 500 N over a distance of 20 m. (2)

1.4 Convert $3{,}409 \times 10^6$ to decimal notation. (1)

1.5 Complete TABLE 1 below by writing only the answer next to the question number (1.5.1–1.5.3) in the ANSWER BOOK.

PHYSICAL QUANTITY	QUANTITY SYMBOL	SI UNIT SYMBOL
work	w	1.5.1
1.5.2	T	K
force	F	1.5.3

TABLE 1

(3 × 1) (3)

[10]

QUESTION 2

2.1 An atom is the smallest particle of an element that has the same properties of the chemical element.

Explain what is meant by an *ionised atom.* (2)

2.2 Draw a simple series circuit consisting of a battery, switch and a resistor. (3)

2.3 The unit of energy was named after James Prescott Joule who was an English physicist.

2.3.1 State *Joule's law*. (3)

2.3.2 Express Joule's law mathematically. (1)

2.4 A 100 W lamp is connected across a 220 V supply.

Calculate the following:

2.4.1 The current flowing through the lamp

2.4.2 The resistance of the filament of the lamp

(2 × 2) (4)

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2.5 Conductors used in the electrical industry are good conductors of electricity.

Name TWO types of materials most commonly used to manufacture these conductors. (2)

2.6 Explain how the following materials affect the resistance of a material:

2.6.1 Cross-sectional area

2.6.2 Length

(2 × 2) (4)

2.7 Calculate the resistance of a copper conductor having a length of 0,716 m. The conductor has a cross-sectional area of 0,043 mm^2 and the resistivity of copper is 1,2 micro-ohm meters. (3)

2.8 Direct current flows in one direction only.

List THREE sources of direct current (DC). (3)

[25]

QUESTION 3

3.1 Refer to FIGURE 1 below and calculate the following:

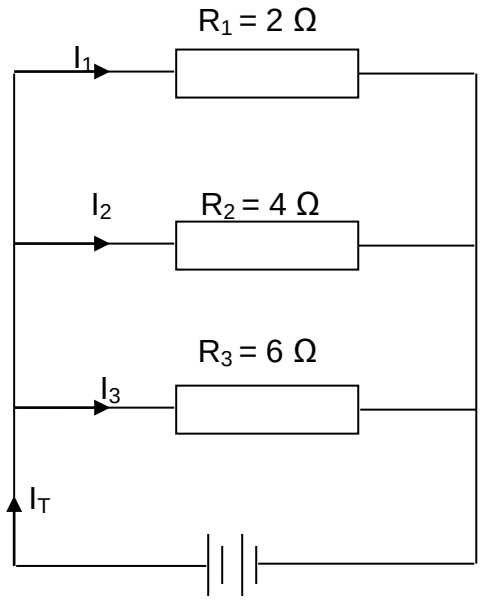


FIGURE 1

3.1.1 The total resistance of the circuit (3)

3.1.2 The currents I_1, I_2 and I_3 (3)

3.1.3 The total current of the circuit (2)

3.2 If the resistor R_1 in FIGURE 1 above is short-circuited, how will it affect the current flowing through resistors R_2 and R_3? (2)

3.3 Explain the following concepts with regard to grouping of electric cells:

3.3.1 Electric cell

3.3.2 Internal resistance

(2 × 2) (4)

3.4 The following load must be connected to a 400 V, three-phase, four-wire supply:

- 3 × 1 kW, 230 V flood lamps
- 6 × 2 kW, 230 V motors
- 9 × 500 W, 230 V lamps

Determine the following:

3.4.1 The total load when all appliances are used simultaneously (4)

3.4.2 The possible load per phase (1)

3.5 Transformers are used to step up or down the voltage in an AC circuit.

Explain what is meant by the following terms:

3.5.1 Turns-ratio

3.5.2 Step-down transformer

(2 × 2) (4)

3.6 You are required to buy a 240/12 V transformer for a project.

Determine the voltage ratio of this transformer. (2)

[25]

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QUESTION 4

4.1 Earthing is required to comply with certain regulations contained in the code of practice.

4.1.1 Explain what is meant by *earthing.* (2)

4.1.2 Explain why earthing of electrical appliances and installations is important. (1)

4.2 An electrician cannot work without measuring instruments.

Explain the difference between *analogue* and *digital* measuring instruments. (4)

4.3 A digital meter is the most commonly used instrument in the electrical field.

List FOUR advantages that digital meters have over analogue meters. (4)

4.4 A galvanometer is used to measure small currents in a DC circuit.

State how the range of this instrument can be extended to measure bigger:

4.4.1 Voltages

4.4.2 Currents

(2 × 1) (2)

4.5 Electricians can use many instruments for measuring and testing.

Name TWO measuring instruments that can be used as continuity testers. (2)

[15]

QUESTION 5

5.1 The secretary at your company requires TWO socket outlets in her office. You were asked to wire this circuit using only ONE circuit breaker.

Show, by means of a neatly labelled circuit diagram, how you would achieve this. (6)

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5.2 TABLE 2 below is a compilation of a parts list for electrical sub-circuits in a domestic installation. Complete this table by writing only the answer next to the question number (5.2.1–5.2.5) in the ANSWER BOOK.

ELECTRICAL SUB-CIRCUIT	SIZE OF WIRE	CIRCUIT BREAKER SIZE
Lights	5.2.1	5.2.2
Socket outlet	5.2.3	5.2.4
5.2.5	6 mm^2	35 A

TABLE 2

(5 × 1) (5)

5.3 Refer to FIGURE 2 below which represents a geyser sub-circuit and identify components A to D. Write only the answer next to the letter (A–D) in the ANSWER BOOK.

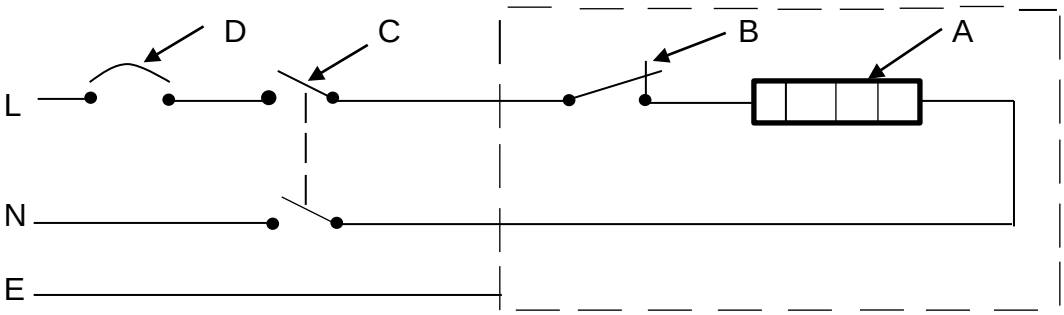


FIGURE 2

(4 × 1) (4)

[15]

QUESTION 6

6.1 FIGURE 3 below indicates components/materials used in the electrical field. Identify the components/materials by writing only the answer next to the question number (6.1.1–6.1.4) in the ANSWER BOOK.



FIGURE 3

(4 × 1) (4)

6.2 You have been contracted by the local municipality to install meter boxes at the dwellings of domestic consumers.

List THREE components that you will find inside a meter box. (3)

6.3 Mercury cells use a reaction between mercuric oxide and zinc electrodes in an alkaline electrolyte.

List THREE practical examples of a mercury cell. (3)

[10]

TOTAL: 100

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FORMULA SHEET – ELECTRICAL PRINCIPLES AND PRACTICE L2

1. $v = \frac{d}{t}$
2. $\bar{\upsilon} = \frac{d}{t}$
3. $F = ma$
4. $S = V_1 I_1 = V_2 I_2$
5. $V_1 = \left(\frac{R_1}{R_t}\right) V_T$
6. $\boldsymbol{I}_1 = \left(\frac{\boldsymbol{R}_2}{\boldsymbol{R}_1 + \boldsymbol{R}_2}\right) \boldsymbol{I}_T$
7. $R_T = R_1 + R_2 + R_3 + \cdots + R_n$
8. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$
9. $R_T = \frac{R_1 R_2}{R_1 + R_2}$
10. $\tau = Fr$
11. $P = \frac{F}{A}$
12. $I_T = I_1 = I_2 = I_3$
13. $I_T = I_1 + I_2 + I_3$
14. $\boldsymbol{R}_{sh} = \frac{\boldsymbol{I}_m \boldsymbol{R}_m}{\boldsymbol{I}_{sh}}$
15. $R_{se} = \frac{V_t}{I_t} - R_m$
16. $F = B\,I\,l$
17. $a = \frac{\Delta v}{\Delta t}$
18. $E = V + I\,r$
19. $\boldsymbol{V} = \boldsymbol{I}\,\boldsymbol{R}$
20. $R = \frac{p\,l}{A}$
21. $\boldsymbol{R}_t = \boldsymbol{R}_0(1 + \boldsymbol{\alpha t})$
22. $W = mg$
23. $P = V\,I$
24. $E = I^2\,R\,t$
25. $w = Fs$
26. $\rho = \frac{m}{V}$
27. $V_T = V_1 + V_2 + V_3$
28. $V_T = V_1 = V_2 = V_3$
29. $Mmf = N\,I$
30. $H = \frac{Mmf}{l}$
31. $B = \frac{\Phi}{A}$
32. $E = B\,l\,v$

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33. $A = \frac{\pi\ d^2}{4}$

34. $I_T = \frac{Emf}{R + r_T}$

35. $E = Pt$

36. $P = \frac{V^2}{R}$

37. $P = \frac{w}{t}$

38. $A = \pi r^2$

39. $f = \frac{1}{t}$

40. $P = I^2R$

41. $\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$

42. $P = \frac{E}{t}$

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